RadeonPilot

Multi-Agent ROCm Inference Operations on a Single AMD Radeon GPU — Private, Local, Evidence-Bound

AMD Hackathon 2026 · Track 2

THE SCENARIO

RadeonPilot turns one Radeon GPU into a self-inspecting inference server. A fleet of Pi agents performs the full ops loop: **diagnose** the ROCm stack, **deploy** the model endpoint, **benchmark** it, **locate** bottlenecks, and **propose** validated optimizations. Every agent generation runs on the Radeon itself — private by construction, aligned with AMD's ROCm.AI agent-workflow direction.

AGENT FLEET (PI AGENT HARNESS 0.83.0)

Agent	Role
EnvDoctor	GPU / ROCm health from rocm-smi & amd-smi telemetry
DeployPlanner	Engine, context, memory-budget and launch-flag plan
BenchmarkRunner	Interprets measured TTFT / throughput records only
OptimAnalyst	Evidence-cited bottleneck analysis and optimizations
OpsCommander	Synthesis + approval-gated action plan

TRACK 2 CAPABILITY COVERAGE

- Local retrieval: BM25 over a ROCm/vLLM tuning corpus with document IDs
- Tool invocation: deterministic GPU probes, real benchmarks, permission gates
- Multi-step planning: parallel specialists + gated commander plan
- Multi-turn memory: SQLite turns, snapshots, audit lineage
- Privacy: loopback-only, disabled Pi tools, redacted audit
- Radeon core inference: attested via amd-smi device occupancy

ARCHITECTURE

```
Gradio Ops Console (127.0.0.1)
   |
Python controller — tools | RAG | memory
   |
Localhost Pi HTTP bridge (Node 22)
   |-- EnvDoctor -- DeployPlanner --
   |-- BenchmarkRunner -- OptimAnalyst --   (parallel)
   v
OpsCommander  ->  gated action plan
   |
OpenAI-compatible endpoint (llama.cpp ROCm)
   v
AMD Radeon gfx1100 · 48 GB · ROCm 7.2.1
```

MEASURED PERFORMANCE (QWEN3-8B, 5-PROMPT SET)

Profile	Mean TTFT	Decode tok/s	VRAM peak
Baseline (F16, default)	**56.2 ms**	**43.6**	17.0 GB
Optimized (F16, FA + 8 slots)	**57.1 ms**	**43.6**	17.0 GB
Quantized (Q4_K_M + FA)	**37.5 ms**	**88.7**	6.5 GB

Quantization doubles decode throughput (43.6 → 88.7 tok/s, 2.03x) and cuts TTFT by 33% with the frozen quality gate passing; FA + parallel slots keep four specialist agents serving concurrently without queueing. All numbers captured on the competition Radeon instance by `scripts/benchmark_vllm.py`; raw samples and amd-smi attestation are committed in `benchmarks/results/`. No fabricated figures.

REPRODUCE IN 4 COMMANDS

`bash scripts/setup_server.sh` ▶ `bash scripts/start_endpoint.sh` ▶ `bash scripts/run_bridge.sh` ▶ `bash scripts/run_app.sh`

Full checkpoint-by-checkpoint guide: `docs/REPRODUCE.md`